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DRAFT REPORT

The Art of Scientific Computation: Respiratory Rates

Estimating Respiratory Rate from ECG Data

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Estimating respiratory information from long-duration electrocardiogram recordings through R-peak detection, R-R interval analysis, and time-series methods.

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Chapter 1

Respiratory Rates

1.1 Introduction

Respiratory rate, defined as the number of complete breathing cycles per minute, is an important physiological vital sign [1]. Accurate measurement of respiratory rate can provide useful information about respiratory health and can be particularly important in the assessment of respiratory illness. For example, pneumonia remains a major cause of mortality among young children worldwide [5]. Accurate respiratory-rate measurement is particularly valuable when direct respiratory monitoring is unavailable or difficult to perform consistently. An automated and reproducible method for estimating respiratory rate from another routinely measured physiological signal may therefore be useful in both clinical and research settings.

This project investigates whether respiratory information can be extracted from electrocardiogram (ECG) recordings. An ECG records the electrical activity of the heart over time and contains a characteristic sequence of features associated with each heartbeat. Of particular interest is the QRS complex and its prominent R peak. Once successive R peaks have been identified, the time interval between them can be calculated.

Breathing can influence heart rate through respiratory sinus arrhythmia. During a breathing cycle, heart rate may increase during inspiration and decrease during expiration. As a result, the time intervals between successive R peaks may contain periodic variation related to respiration. This provides a possible route for estimating respiratory rate indirectly from ECG data.

The main research question addressed by this project is:

Can respiratory rate be estimated from variations in the R–R intervals of long-duration ECG recordings?

The project also considers several related questions:

- How reliably can R peaks be detected from ECG recordings with varying morphology and noise?
- How should the R–R intervals be represented and processed before respiratory analysis?
- Can frequency-domain or time-series methods identify a respiratory-related periodic component in the R–R interval sequence?
- How stable is the resulting respiratory-rate estimate over time?
- Are there observable differences between normal and apnea-labelled intervals?

The primary goal is to construct a transparent and reproducible computational pipeline from the raw ECG waveform to a respiratory-rate estimate. Automated classification of sleep apnea is treated as a possible extension rather than the main objective of the project.

1.2 Background and Theory

1.2.1 Electrocardiogram and the QRS Complex

An electrocardiogram measures electrical activity generated by the heart. A typical cardiac cycle contains a P wave, QRS complex, and T wave. The QRS complex is associated with ventricular depolarisation and usually contains a prominent R peak. Because the R peak is often the most easily identifiable feature of the ECG waveform, it provides a useful timing marker for successive heartbeats.

If the detected R-peak times are written as

$$t_1, t_2, \dots, t_N, \quad (1.1)$$

then the corresponding R–R intervals are

$$RR_i = t_{i+1} - t_i. \quad (1.2)$$

The R–R interval is therefore the time between two consecutive heartbeats. Changes in this interval reflect changes in instantaneous heart rate.

A major challenge is that ECG morphology can vary both between subjects and within the same recording. Electrode placement, movement artefacts, noise, ectopic beats, and physiological variation can all affect the waveform. Consequently, a peak-detection method based on a fixed absolute threshold may perform well for one record but fail for another.

1.2.2 Respiratory Sinus Arrhythmia

Respiratory sinus arrhythmia describes a relationship between respiration and heart rate. In many individuals, heart rate increases slightly during inspiration and decreases during expiration. This causes the R–R intervals to vary over the breathing cycle.

If this respiratory modulation is sufficiently strong, the R–R interval sequence can be treated as an ECG-derived respiratory signal. A periodic component in this signal may then provide an estimate of respiratory rate.

However, R–R variability is not caused only by respiration. Other physiological processes, measurement noise, missed R peaks, falsely detected peaks, and ectopic beats can also modify the interval sequence. The central challenge of the project is therefore not simply to find a periodic frequency, but to determine whether a detected periodic component is plausibly associated with respiration.

1.2.3 Frequency-Domain Analysis

One possible method for estimating the dominant periodic component is the discrete Fourier transform. For a uniformly sampled time series $x_0, x_1, \dots, x_{N-1}$, the discrete Fourier transform is

$$X_k = \sum_{n=0}^{N-1} x_n \exp\left(-\frac{2\pi i k n}{N}\right). \quad (1.3)$$

If the sampling frequency of the processed signal is f_s , then the frequency corresponding to Fourier coefficient k is

$$f_k = \frac{k f_s}{N}. \quad (1.4)$$

A dominant spectral peak within a physiologically plausible respiratory frequency range may provide an estimate of breathing frequency, f_{resp} . This can be converted from hertz to breaths per minute using

$$R_{\text{resp}} = 60 f_{\text{resp}}, \quad (1.5)$$

where R_{resp} is the estimated respiratory rate in breaths per minute.

A complication is that R–R intervals are naturally irregularly sampled because their timestamps are determined by the heartbeats themselves. Therefore, direct application of a standard FFT may require interpolation onto a uniformly sampled time grid. The effects of this preprocessing must be considered when interpreting the resulting spectrum.

1.3 Data

The project uses data from the publicly available Apnea-ECG database [2]. The dataset contains long-duration single-channel ECG recordings collected as part of a sleep-apnea study.

Several file types are relevant to the analysis:

File type	Purpose
.dat	ECG waveform data
.hea	Record metadata and signal information
.qrs	QRS-complex annotations
.apn	Sleep-apnea annotations

Table 1.1: Main file types used from the Apnea-ECG dataset.

The waveform and header files provide the ECG signal and the information required to interpret it. Where available, the QRS annotations provide reference locations for QRS complexes, while the apnea annotations identify periods labelled as normal or apneic.

The WFDB annotation format is more complex than a simple text format and contains compact binary annotation records. The WFDB specification [3] is therefore used as a reference when reading or interpreting annotation files.

Not every record contains the same set of annotations, and the available QRS annotations should not automatically be treated as perfectly accurate. They are instead used as a useful reference for testing the ECG-processing pipeline.

1.4 Methods

The planned analysis follows the computational pipeline

$$\text{ECG} \rightarrow \text{R peaks} \rightarrow \text{R–R intervals} \rightarrow \text{respiratory-related signal} \rightarrow \text{respiratory-rate estimate.} \quad (1.6)$$

1.4.1 ECG Loading and Visualisation

The first stage is to load the ECG waveform and associated metadata and convert the signal from sample number to physical time. Representative segments of the ECG are then visualised together with the available QRS and apnea annotations.

This stage serves two purposes. First, it verifies that the data and annotations have been interpreted correctly. Second, visual inspection helps identify differences in morphology, noise level, and signal amplitude that may affect later R-peak detection.

The initial objective is to produce a figure containing a short ECG interval, the corresponding QRS locations, and any overlapping apnea annotation. This provides a baseline visualisation against which the later automated analysis can be checked.

1.4.2 R-Peak Detection

The next stage is to identify the R peaks directly from the ECG waveform. The detector must be sufficiently robust to deal with changes in signal amplitude and morphology.

A simple fixed-amplitude threshold is unlikely to generalise across all records. The detection method will therefore use relative signal properties such as local peak prominence, minimum peak

separation, or an adaptive threshold. A minimum temporal separation between candidate peaks can also be used to prevent multiple detections within a single QRS complex.

Where QRS annotations are available, detected R peaks will be compared with the annotation locations. The comparison will be used to identify common failure modes such as missed peaks, false peaks, and timing offsets.

Previous work has shown that the choice of QRS-detection algorithm can affect derived heart-rate variability measurements, making this validation an important part of the analysis [4].

1.4.3 R–R Interval Construction

After detecting the R peaks, the intervals are calculated using Equation 1.2. Each interval is associated with a representative time, for example the timestamp of the later R peak or the midpoint between the two peaks.

Before respiratory analysis, implausible R–R intervals must be identified. Abnormally short or long intervals may be caused by missed detections, false detections, ectopic beats, or noise. Such intervals can strongly influence a frequency-domain analysis.

The resulting R–R sequence is irregularly sampled. If a conventional FFT is used, the cleaned interval series will therefore be interpolated onto a regular time grid. The interpolation frequency must be high enough to capture respiratory variation without unnecessarily increasing the amount of data.

1.4.4 Respiratory-Rate Estimation

The respiratory-rate estimation stage will analyse periodic variation in the processed R–R interval series.

The baseline approach is:

1. detect R peaks from the ECG;
2. calculate the R–R intervals;
3. remove or correct clearly implausible intervals;
4. interpolate the R–R sequence onto a regular time grid;
5. remove slow trends where necessary;
6. divide the signal into analysis windows;
7. calculate the frequency spectrum within each window;
8. identify the dominant plausible respiratory frequency;
9. convert the frequency to breaths per minute using Equation 1.5.

Using multiple time windows rather than a single spectrum of the entire recording should make it possible to investigate how the estimated respiratory rate changes over time.

The choice of analysis-window length introduces a time-frequency trade-off. Longer windows improve frequency resolution but reduce the ability to track rapid changes. Shorter windows provide better temporal localisation but produce less precise frequency estimates.

1.5 Preliminary Results

At the current draft stage, the project is focused on establishing and validating the data-processing pipeline before drawing conclusions about respiratory rate.

The first milestone is successful loading and visualisation of representative ECG records together with their available QRS and apnea annotations. This provides a direct check that the waveform timing and annotation timing are consistent.

The next preliminary result will be a comparison between automatically detected R peaks and the provided QRS annotations. This comparison will be used to determine whether the peak-detection method is sufficiently reliable for construction of the R–R interval series.

Once the R–R series has been validated, representative sections of the interval sequence will be examined in both the time and frequency domains. The main quantity of interest will be the presence and stability of a dominant respiratory-related periodic component.

1.6 Discussion and Current Limitations

Several limitations must be considered when interpreting ECG-derived respiratory estimates.

First, respiratory sinus arrhythmia is not the only source of heart-rate variability. Changes in autonomic activity, ectopic beats, movement artefacts, and other physiological processes may also affect the R–R intervals.

Second, errors in R-peak detection propagate directly into the R–R interval series. A single missed peak may generate an interval approximately twice as long as expected, while an additional false peak may create two artificially short intervals. Reliable peak detection and outlier handling are therefore essential.

Third, interpolation changes the original irregular R–R data and may introduce additional assumptions into the analysis. The interpolation method and resampling rate should therefore be chosen carefully.

A further limitation is the absence of a continuous direct respiratory-rate reference signal in the current project setup. The apnea annotations indicate periods of sleep apnea, but they are not equivalent to point-by-point ground-truth respiratory-rate measurements. Consequently, a spectral peak that appears physiologically plausible should not automatically be interpreted as a fully validated respiratory-rate measurement.

Finally, ECG morphology varies between subjects and recording conditions. A method that performs well on a single record may therefore not generalise to the full dataset.

1.7 Planned Next Steps

The next stages of the project are:

1. complete and validate the ECG and annotation loading pipeline;
2. implement robust R-peak detection and compare the results with available QRS annotations;
3. construct and clean the R–R interval time series;
4. investigate interpolation and preprocessing choices for the irregularly sampled R–R sequence;
5. implement a baseline spectral respiratory-rate estimator;
6. evaluate the stability of respiratory-rate estimates across multiple time windows and ECG records;
7. compare respiratory-related features between normal and apnea-labelled intervals;
8. if time permits, investigate whether simple statistical features can help distinguish normal and apnea-labelled intervals.

The priority remains the development of a reliable and interpretable respiratory-rate estimation pipeline. More complex classification or machine-learning methods will only be considered after the baseline analysis has been validated.

1.8 Conclusion

This project investigates whether respiratory information can be extracted indirectly from long-duration ECG recordings. The proposed approach uses the timing of R peaks to construct an R–R interval time series and then searches for periodic variation associated with respiration.

The central computational challenge is to separate respiratory-related variation from errors in peak detection, normal heart-rate variability, artefacts, and other physiological effects. The draft therefore focuses on building and validating each stage of the processing pipeline before interpreting the final respiratory-rate estimate.

The next phase of the project will determine whether the resulting R–R-derived signal contains a sufficiently stable respiratory component to support meaningful estimates over time and whether these features differ between normal and apnea-labelled intervals.

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